II-I CRT Syllabus (2026-27)

MODULE 1 – Introduction to Python

Topics

- Introduction to Python
- Features of Python
- Applications of Python
- Installation & Environment Setup
- Python IDEs (VS Code, PyCharm, Jupyter Notebook)
- Writing the First Python Program
- Python Execution Process
- Python Syntax
- Comments
- Indentation
- Input & Output Statements

MODULE 2 – Python Fundamentals

Topics

- Variables
- Identifiers
- Keywords
- Constants
- Data Types
    - Numeric
    - Boolean
    - String
    - List
    - Tuple
    - Set
    - Dictionary
- Type Conversion
- Type Casting
- Operators
    - Arithmetic
    - Relational
    - Logical
    - Assignment
    - Membership
    - Identity
    - Expressions

MODULE 3 – Control Flow Statements

Topics

- if Statement

- if-else
- Nested if
- elif Ladder
- match-case
- while Loop
- for Loop
- Nested Loops
- break
- continue
- pass

Practical Programs

- Prime Number
- Armstrong Number
- Palindrome
- Fibonacci Series
- Factorial
- Number Patterns
- Star Patterns

MODULE 4 – Functions & Modules

Topics

- Function Definition
- Function Calling
- Parameters & Arguments
- Return Statement
- Recursive Functions
- Lambda Functions
- Anonymous Functions
- Variable Scope
- Global & Local Variables
- Modules
- Packages
- Import Statements

MODULE 5 – Python Collections

Topics

Lists

- Creation
- Indexing
- Slicing
- List Methods
- Nested Lists
- List Comprehensions

Tuples

- Creation
- Packing & Unpacking
- Tuple Operations

Sets

- Set Operations

- Set Methods
- Frozen Sets

Dictionaries

- Dictionary Methods
- Nested Dictionaries
- Dictionary Comprehensions

MODULE 6 – Strings & Regular Expressions

Topics

- String Operations
- Indexing
- Slicing
- String Methods
- String Formatting
- Regular Expressions (re Module)
- Pattern Matching

MODULE 7 – File Handling & Exception Handling

File Handling

- File Operations
- Reading Files
- Writing Files
- Appending Files
- CSV Files
- JSON Files

Exception Handling

- try
- except
- else
- finally
- raise
- Custom Exceptions

MODULE 8 – Object-Oriented Programming

Topics

- Classes
- Objects
- Constructors
- Instance Variables
- Class Variables
- Methods
- Encapsulation
- Inheritance

- Polymorphism
- Method Overloading
- Method Overriding
- Abstraction
- Magic Methods

MODULE 10 – Algorithm Analysis

Topics

- Algorithm Design
- Time Complexity
- Space Complexity
- Asymptotic Notations
    - Big-O
    - Omega (Ω)
    - Theta (Θ)

      Best, Average & Worst Case Analysis

MODULE 11 – Linear Data Structures

Arrays

- Operations
- Traversal
- Insertion
- Deletion
- Searching

Strings

- String Algorithms
- Pattern Matching

Linked Lists

- Singly Linked List
- Doubly Linked List
- Circular Linked List
- Operations

Stacks

- Stack Operations
- Stack Applications
- Expression Evaluation

Queues

- Simple Queue
- Circular Queue
- Deque
- Priority Queue
- Queue Applications

MODULE 12 – Searching & Sorting Algorithms

Searching

- Linear Search
- Binary Search

Sorting

- Bubble Sort
- Selection Sort
- Insertion Sort
- Merge Sort
- Quick Sort
- Heap Sort
- Counting Sort (Introduction)